



INTERMEDIATE ANALITICS

Himalayan Expedition

A 100-Year Analysis of Himalayan Expeditions

TEAMS MEMBERS



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**Visualization Design & Narrative
Synthesis**



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Presentation Flow

Mountaineering Data Analysis: Risks, Realities, and Revelations

- 1 The Climber's Glossary**
Understanding Key Mountaineering Terms
- 2 System Architecture & Tech Stack**
Behind the Scenes: Data Engineering Overview
- 3 SQL Introduction**
Preparing for Live Analysis and Core Queries
- 4 Live Data Analysis & Visualizations**
Peak Danger, Demographics, Guides, Commercialization, and Global Patterns

Review of Findings and Exploring the Codebase

The Climber's Glossary

Key Terminology in Mountaineering



The Death Zone

Altitudes above 8,000m where the human body cannot acclimatize and oxygen is vital for survival.



Hired / Sherpa

High-altitude professionals providing logistical support. Our data reveals they face significantly higher risks than clients.



Lethality Index

A statistical measure of a peak's danger, calculated as the ratio of deaths to successful summits (e.g., Annapurna's 1-in-7 risk).



Commercialization

The shift from elite, small-team exploration to large-scale, agency-led 'mountain tourism'.

System Architecture & Tech Stack

Showcasing the engineering "under the hood"



Cloud Host

Azure SQL Database



ETL/Batching

Python scripts for data processing



Automation

PowerShell for task automation



Database Design

3rd Normal Form (3NF) applied, decomposing data into 14 relational tables to enhance efficiency and reduce redundancy.



Query Techniques

Utilized advanced JOINS, CTEs, Window Functions for ranking, and CASE Statements for risk classification.



Core SQL Blocks

The Storytelling Blocks: SQL Introduction

Our live query demo is built upon extracting key narratives from data using four core SQL blocks. These scripts will lay the foundation for the visualizations you're about to see:

- **04-death-zone-demographics.sql**: Analyzes the human cost and peak lethality.
- **05-sherpa-indispensable.sql**: Explores labor disparity and mortality risks for guides.
- **06-everest-effect.sql**: Traces the 100-year explosion of commercial climbing.
- **07-national-cultures.sql**: Examines global participation and success patterns.

Transition to Live Analysis

Final prompt to switch screens for live data review.



- **GitHub Codebase Review**

Pivot to the repository to examine the T-SQL logic and ETL scripts.

- **Visual Evidence**

Conclude with the 4 key visualizations proving data findings on peak danger and guide mortality.

This concludes our initial findings. We will now transition to the live environment for further analysis.



The Evolving Landscape of Everest

Beyond the Summit

The Transformation: Himalayan climbing has evolved from elite exploration to mass-market tourism.

The Persistent Threat: Despite modern advancements, the "Death Zone" remains unforgiving.

The Hidden Burden: Local guides absorb the most risk, facing a 2-3x higher mortality rate than their clients.

A Globalized Ecosystem: Western nations drive economic demand, while Nepal provides indispensable human infrastructure.